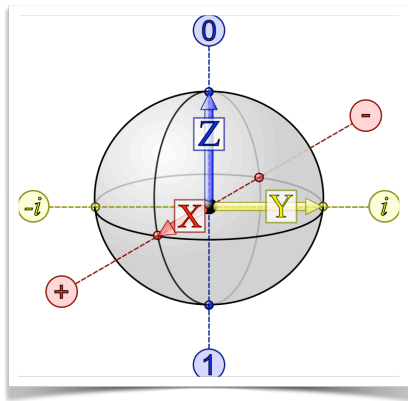




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IQC 2024/25 (756AA, 6CFU)

Introduction to Quantum Computing

The postulates of Quantum Mechanics

An Introduction to Quantum Computing
by Phillip Kaye, Raymond Laflamme, Michele Mosca

Quantum Computing for Computer Scientist
by Noson S. Yanofsky, Mirco A. Mannucci

Quantum mechanics distilled
by Andy Matuschak and Michael Nielsen
<https://quantum.country/qm>
San Francisco (2020)

Foundations of Quantum Programming
by Mingsheng Ying

What is quantum mechanics?

- **Quantum mechanics is a set of four postulates that provide a mathematical framework for describing the behaviour of physical systems** (the universe and everything in it)
- **Quantum mechanics is NOT a fully specified physical theory.** It's a framework to construct other physical theories
- We can think of **quantum mechanics** as analogous to an **operating system** for a computer
 - The operating system accomplishes important basic functions
 - Users need another layer of application software on top of those basic functions
 - The application layer runs within the framework provided by the operating system, but it is not a part of the operating system
 - The **application layer** is analogous to **physical theories**

Four postulates

Specify a general framework for describing the behaviour of a physical system:

1. **Statics** - How to describe the state of a **closed** quantum system
2. **Dynamics** - How to describe the evolution of a **closed** quantum system, i.e., how it is allowed to change with time
3. **Measurement** - How information is extracted from a closed quantum system via interactions with an external system
4. **Composite systems** - How to describe the state of a composite system in terms of its component parts

A **closed** (or isolated) system is an ideal physical system that does not interact at all with its environment

The first postulate: state space

Postulate 1: State Space Postulate

Associated to any physical system is a **complex Hilbert space** known as the **state space** of the system.

If the system is *isolated*, then the system is completely described by its **state vector**, which is a **unit vector** in the its state space.

A Hilbert space is a **vector space** equipped with an **inner product**

Note

- The postulate tells us that **every physical system has a state space** (an atom has a state space; a human being has a state space; even the universe as a whole has a state space)
 - but *it does not tell us what the state space is*, for any given physical system
 - that needs to be figured out on a case-by-case basis, with additional physical theories
- Different types of physical systems have different state spaces
- Depending on the system considered, the Hilbert space may be infinite-dimensional
- We will always assume that the state is described by a vector in a **finite-dimensional complex vector space**

State of a qubit

- A **qubit** is the simplest of all isolated quantum systems
- For a qubit the state space is the **two-dimensional complex vector space $\mathbb{C}^2$**
- The state of a qubit is described by a **unit vector in $\mathbb{C}^2$**
- Any physical system whose state space can be described by $\mathbb{C}^2$ can be used to implement a qubit. Examples of physical realization of qubits:
 - spin of an electron
 - polarization of a photon
 - current in a superconducting circuit

- The general state of a qubit is expressed by the vector

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha, \beta \in \mathbb{C}$$

α e β are called the **amplitudes** of the basis states $|0\rangle$ and $|1\rangle$, respectively

- The condition that the state vector is a unit vector (**Normalization constraint**) means that

$$\langle\psi|\psi\rangle = |\alpha|^2 + |\beta|^2 = 1$$

Bloch sphere

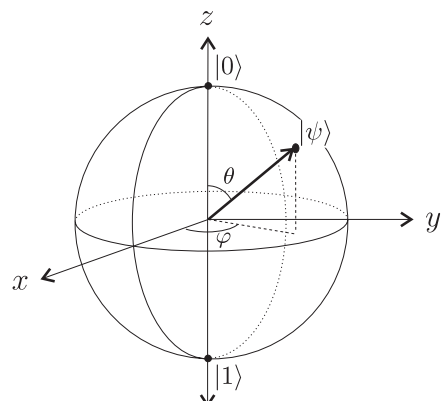
- Qubit states can be geometrically represented as **point on the surface of a sphere of unitary radius** known as **Bloch's sphere**
geometrical representation (and visualization) of a qubit in a **3D space**
- Many operations on **single** qubits can be neatly described within the Bloch sphere picture (but there is no simple generalization of the Bloch sphere known for multiple qubits)

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha, \beta \in \mathbb{C}$$

$$|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\varphi}\sin(\theta/2)|1\rangle$$

$$0 \leq \theta \leq \pi$$

$$0 \leq \varphi \leq 2\pi$$



The second postulate: evolution

Postulate 2: Evolution Postulate

The time evolution of a closed quantum system is described by the **Schroedinger equation**

$$i \hbar \frac{d|\psi(t)\rangle}{dt} = H(t) |\psi(t)\rangle$$

where $\hbar$ is a physical constant (Planck's constant) and $H(t)$ is a fixed Hermitian operator, known as the **Hamiltonian of the system**

Note

- The postulate describes the evolution of a quantum system in continuous time, with the change described by a differential equation
- The **Hamiltonian $H(t)$** represents the **total energy function** for the system and 'tells' the quantum state how to change
- If we know the Hamiltonian of a system, then we understand its dynamics completely, at least in principle.
- In general figuring out the Hamiltonian needed to describe a particular physical system is a very difficult problem

The second postulate simplified: discrete time evolution

In the quantum circuit contexts we have discrete gates being applied to cause discrete changes in the quantum state

- We will consider only discrete computational steps
- The postulate can be simplified (and we will not need to discuss Hamiltonians)

Postulate 2': Evolution Postulate

The **discrete** time evolution of a closed quantum system is described by a **unitary transformation**:

The state $|\psi_1\rangle$ of the system at a time t_1 is related to the state $|\psi_2\rangle$ of the system at a later time t_2 by a **unitary matrix U** which depends only on the times t_1 and t_2 ,

$$|\psi_2\rangle = U|\psi_1\rangle$$

An **invertible complex square matrix U** is **unitary** if its **matrix inverse U^{-1}** equals its **conjugate transpose $U^\dagger$**

$$U^\dagger U = U U^\dagger = I$$

The second postulate simplified: discrete time evolution

Note

- This expression follows directly from the Schroedinger equation:
every unitary operator U can be realized as a solution of Schroedinger's equation
- *The postulate does not tell us which unitary transformations describe real-world quantum dynamics:* they need to be figured out case by case.
- Why unitary matrices?

Unitary matrices are the only matrices which **preserve length**

And we want the quantum state to remain normalized, so that the probabilities of measurement outcomes sum to one

The Pauli matrices

The **Pauli matrices** X , Y and Z are important one-qubit 2×2 unitary matrices

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

The Pauli matrices perform 180-degree rotations around their corresponding axes.

Each Pauli matrix P is both **unitary** and **Hermitian**, and hence **self-inverse**

$$P^{-1} = P$$

Effect on the computational basis states

bit-flip operator	$X 0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 1\rangle$	$X 1\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 0\rangle$
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bit- and phase-flip operator	$Y 0\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ i \end{bmatrix} = i 1\rangle$	$Y 1\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -i \\ 0 \end{bmatrix} = -i 0\rangle$
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phase-flip operator	$Z 0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 0\rangle$	$Z 1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = - 1\rangle$
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Effect of unitary matrices on superposition of basis states

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\xrightarrow{U}$$

$$U|\psi\rangle = U(\alpha|0\rangle + \beta|1\rangle) = \alpha U|0\rangle + \beta U|1\rangle$$

Effect of Pauli gates on superposition of basis states:

The Pauli-X gate swaps the amplitudes of $|0\rangle$ and $|1\rangle$

$$\alpha|0\rangle + \beta|1\rangle \longrightarrow \boxed{\mathbf{X}} \longrightarrow \beta|0\rangle + \alpha|1\rangle \qquad X \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \beta \\ \alpha \end{bmatrix}$$

The Pauli-Y gate swaps the amplitudes of $|0\rangle$ and $|1\rangle$, multiplies each amplitude by i , and negates the amplitude of $|1\rangle$

$$\alpha|0\rangle + \beta|1\rangle \longrightarrow \boxed{\mathbf{Y}} \longrightarrow -i\beta|0\rangle + i\alpha|1\rangle \qquad Y \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} -i\beta \\ i\alpha \end{bmatrix}$$

The Pauli-Z gate negates the amplitude of $|1\rangle$, leaving the amplitude of $|0\rangle$ the same

$$\alpha|0\rangle + \beta|1\rangle \longrightarrow \boxed{\mathbf{Z}} \longrightarrow \alpha|0\rangle - \beta|1\rangle \qquad Z \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ -\beta \end{bmatrix}$$

The Hadamard matrix

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

It is both **unitary** and **Hermitian**, and hence **self-inverse**: $H^{-1} = H$

H puts the computational basis states in **uniform superposition**:

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$$

$$H|1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle$$

Since H is self-inverse, it interferes the superposition to recover the original computational basis states

$$H|+\rangle = |0\rangle \quad H|-\rangle = |1\rangle$$

Z basis
Computational basis

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

X basis
Hadamard basis

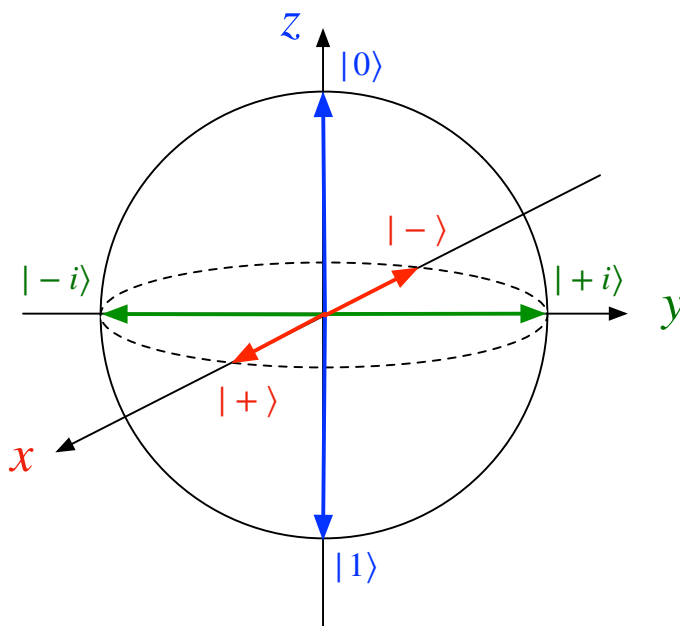
$$|+\rangle = H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$|-\rangle = H|1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Y basis
Circular basis

$$|+i\rangle = S|+\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ i \end{bmatrix} = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}$$

$$|-i\rangle = S|-\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -i \end{bmatrix} = \frac{|0\rangle - i|1\rangle}{\sqrt{2}}$$



$$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$$

Phase gate
changes the phase of $|1\rangle$
by a factor of i

Measurements

- ▶ Measurement corresponds to **transforming the quantum information** (stored in a quantum system) **into classical information**
- ▶ For example, measuring a qubit typically corresponds to reading out a classical bit, i.e., whether the qubit is 0 or 1
- ▶ A central principle of quantum mechanics is that **measurement outcomes are probabilistic**
 - Only the probability of observing specific values can be calculated
 - measurement is a nondeterministic process
 - measurement outcomes are probabilistic
 - Quantum systems get perturbed and modified as a result of measuring them

Measurements

- ▶ **Closed** quantum systems evolve according to a unitary operator
- ▶ In order to **observe** and **measure** some properties of a system

*we must allow the system to **interact** with the measurement apparatus of some observer (an external physical system)*
- ▶ When this happens *the original system is no longer closed*, and the *Evolution Postulate is no longer appropriate* for describing its evolution:

*the evolution of the state of a system during a measurement process is **not unitary***

The third postulate provides a **mathematical formalism for measurements**

- It tells us how to compute
 - ▶ the **probability $p(m)$ that outcome m occurs**,
 - ▶ and **the new state of the system** after the measurement with outcome m
- It does not tell us what measurement can be done in practice (some measurements can be simple to state mathematically, but not easy to implement)

The third postulate: Measurement

Postulate 3: Measurement Postulate

A Quantum measurement on a quantum system has some set of outcomes and is described by a collection of *measurement operators* M_m - one operator for each possible outcome m

The index m labels the measurement outcomes that may occur in the experiment

The measurement operators are linear (not unitary) operators acting on the state space of the system being measured, that satisfy the *completeness relation*

$$\sum_m M_m^\dagger M_m = I$$

If the state of the quantum system is $|\psi\rangle$ immediately before the measurement, then for each m

- the probability that result m occurs is given by

$$p(m) = \langle \psi | M_m^\dagger M_m | \psi \rangle = \| M_m | \psi \rangle \|^2$$

- and the state of the system after the measurement with outcome m is

$$\frac{M_m | \psi \rangle}{\sqrt{\langle \psi | M_m^\dagger M_m | \psi \rangle}}$$

The third postulate: Measurement

- This is only a mathematical formalism for measurements
- The state of the system after the measurement

$$\frac{M_m | \psi \rangle}{\sqrt{\langle \psi | M_m^\dagger M_m | \psi \rangle}} = \frac{M_m | \psi \rangle}{\sqrt{p(m)}}$$

is a properly **normalized quantum state**, and in general it is **NOT** a scalar multiple of $|\psi\rangle$: *the measurement has modified the state of the system*

- The denominator vanishes only if $p(m) = 0$ (this outcome will never occur)
- The **completeness equation** expresses the fact that **probabilities sum to one**

$$\sum_m p(m) = \sum_m \langle \psi | M_m^\dagger M_m | \psi \rangle = \langle \psi | I | \psi \rangle = \langle \psi | \psi \rangle = 1$$

($|\psi\rangle$ is arbitrary and not dependent on the index m)

Global phase factor

Suppose we have two quantum states related by a global phase factor

$$|\psi\rangle \text{ and } |\psi'\rangle = e^{i\theta} |\psi\rangle$$

The measurement probabilities for the two states are always **identical**

$$\begin{aligned} p'(m) &= \langle \psi' | M_m^\dagger M_m | \psi' \rangle \\ &= e^{-i\theta} \langle \psi | M_m^\dagger M_m e^{i\theta} | \psi \rangle \\ &= \langle \psi | M_m^\dagger M_m | \psi \rangle = p(m) \end{aligned}$$

The statistics of any measurement we could perform on the state $|\psi'\rangle = e^{i\theta} |\psi\rangle$ are exactly the same as they would be for the state $|\psi\rangle$

- ➡ Global phases have no observable consequences
- ➡ This explains the claim that global phases have no physical significance

Projective measurements

- For many applications of quantum computation and quantum information, we will mostly perform **PROJECTIVE MEASUREMENTS**
- For these measurements, the measurement operators are **projectors**

Projectors are operators P which satisfy $P^2 = P$

An example of projector operators is the operator $|0\rangle\langle 0| = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

$$(|0\rangle\langle 0|)^2 = |0\rangle\langle 0| |0\rangle\langle 0| = |0\rangle \underbrace{\langle 0|0\rangle}_1 \langle 0| = |0\rangle\langle 0|$$

- The **measurement operators** can be derived from the spectral decomposition of a Hermitian operator M associated with the measurement

An operator M is **Hermitian** if it is self-adjoint: $M^\dagger = M$
(it is equal to its own conjugate transpose)

In physics, Hermitian operators are called **OBSERVABLES**

- An **observable** is a *property of a physical system, i.e., a physical quantity that can be measured* (e.g., position, polarization, etc.)
- Each **physical observable** corresponds to a **Hermitian operator** M
- All eigenvalues of an observable are **real**, and the eigenvectors are **orthogonal**
- The possible **outcomes of a measurement** correspond to the **eigenvalues** of the **observable** M

Observables can be thought of as questions we can pose to quantum systems

Each question admits a set of answers: the eigenvalues of the observable

The **measurement operators** can be derived from the **spectral decomposition** of the **observable** M

Any observable M can be written as

$$M = \sum_m m |u_m\rangle\langle u_m| = \sum_m m P_m \quad \text{Spectral decomposition}$$

where

- m are the real eigenvalues of M (*outcomes of the measurement*)
- $|u_m\rangle$ is the eigenvector associated to m
- $P_m = |u_m\rangle\langle u_m|$ is the **projector** onto the eigenspace of M with real eigenvalue m , and is the **measurement operator** associated to the outcome m (i.e., $M_m = P_m = |u_m\rangle\langle u_m|$)

The third postulate: Projective measurements

- The possible **outcomes of the measurement** correspond to the **eigenvalues** m of the **observable**

$$M = \sum_m m P_m = \sum_m m |u_m\rangle\langle u_m|$$

- Upon measuring the state $|\psi\rangle$, the probability of getting result m is given by

$$p(m) = \langle\psi|P_m^\dagger P_m|\psi\rangle = \langle\psi|P_m|\psi\rangle$$

- Given that outcome m occurred, the state of the quantum system immediately after the measurement is

$$\frac{P_m|\psi\rangle}{\sqrt{p(m)}}$$

- The operators P_m satisfy the completeness relation

$$\sum_m P_m^\dagger P_m = \sum_m P_m = I$$

Projective measurements

Since all possible outcomes of a projective measurement are real numbers, we can compute the **expectation** – average value – of M in state $|\psi\rangle$

$$\begin{aligned} E_\psi[M] &= \sum_m m p(m) \\ &= \sum_m m \langle\psi|P_m|\psi\rangle \\ &= \langle\psi|\sum_m m P_m|\psi\rangle \\ &= \langle\psi|M|\psi\rangle \end{aligned}$$

Measurement in the standard basis

In computer science, by measurement we often mean **single qubit measurement in the computational $|0\rangle, |1\rangle$ basis**

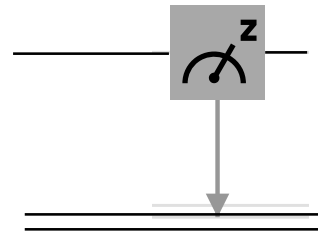
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

The outcome of the measurement is

- **0** with probability $p(0) = |\alpha|^2$
- **1** with probability $p(1) = |\beta|^2$

The state of the system after this measurement is now

- **$|0\rangle$** with probability $|\alpha|^2$
- **$|1\rangle$** with probability $|\beta|^2$



Born rule

This measurement has two outcomes (0 and 1) and corresponds to perform a measurement with **measurement operators**

$$P_0 = |0\rangle\langle 0| = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$P_1 = |1\rangle\langle 1| = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

The completeness equation is satisfied

$$\sum_m P_m^\dagger P_m = \sum_m P_m = P_0 + P_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = I$$

The measurement operators are **projectors onto $|0\rangle$ and $|1\rangle$, respectively,**

$$P_0^2 = P_0$$

$$P_1^2 = P_1$$

Measurements in the computational basis are **projective measurements**

$$P_0 = |0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$P_1 = |1\rangle\langle 1| = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Assume our system is in the state

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

According to the third postulate, the probability of obtaining the outcome 0 is

$$p(0) = \langle \psi | P_0^\dagger P_0 | \psi \rangle = \langle \psi | P_0 | \psi \rangle = \langle \psi | 0 \rangle \langle 0 | \psi \rangle = \alpha^* \alpha = |\alpha|^2$$

Similarly, the probability of obtaining the measurement outcome 1 is

$$p(1) = \langle \psi | P_1^\dagger P_1 | \psi \rangle = \langle \psi | P_1 | \psi \rangle = \langle \psi | 1 \rangle \langle 1 | \psi \rangle = \beta^* \beta = |\beta|^2$$

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

The state after measurement in the two cases is

$$\frac{P_0 |\psi\rangle}{\sqrt{p(0)}} = \frac{|0\rangle\langle 0 | \psi \rangle}{|\alpha|} = \frac{\alpha}{|\alpha|} |0\rangle \equiv |0\rangle$$

$$\frac{P_1 |\psi\rangle}{\sqrt{p(1)}} = \frac{|1\rangle\langle 1 | \psi \rangle}{|\beta|} = \frac{\beta}{|\beta|} |1\rangle \equiv |1\rangle$$

The factors $\alpha/|\alpha|$ and $\beta/|\beta|$ are global phase factors

$$\alpha = |\alpha| e^{i\phi}, \text{ for same phase angle } \phi \in [0, 2\pi) \rightarrow \frac{\alpha}{|\alpha|} = e^{i\phi}$$

$$\beta = |\beta| e^{i\gamma}, \text{ for same phase angle } \gamma \in [0, 2\pi) \rightarrow \frac{\beta}{|\beta|} = e^{i\gamma}$$

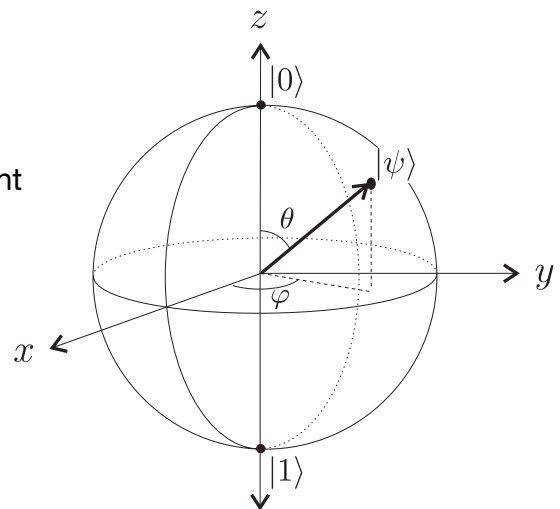
Thus, these states can be regarded as **equivalent to the quantum states** $|0\rangle$ and $|1\rangle$, respectively, as expected for a measurement in the computational basis

Measurement in the standard basis

Observable?

It can be viewed as a projective measurement with respect to the Pauli gate Z

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$



The Pauli Z gate is a Hermitian operator and can be interpreted as an observable corresponding to the measurement of the projection of a qubit's state vector along the z-axis.

Measurement in the standard basis

Z has eigenvalues +1 and -1 with corresponding eigenvectors $|0\rangle$ and $|1\rangle$, and its spectral decomposition is

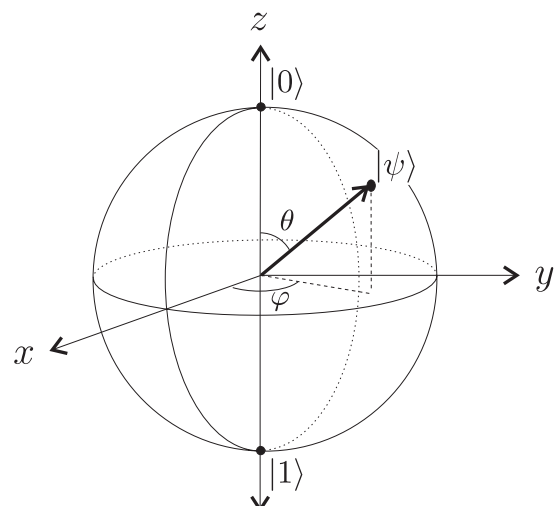
$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \underbrace{|0\rangle\langle 0|}_{P_0} - \underbrace{|1\rangle\langle 1|}_{P_1}$$

The measurement of Z on the state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

gives

- the result +1 with probability $|\alpha|^2$, and the state after the measurement is $|0\rangle$
- the result -1 with probability $|\beta|^2$, and the state after the measurement is $|1\rangle$



Measurement in an orthonormal basis

A measurement in an orthonormal basis $\{ |m\rangle \}$ corresponds to perform a **projective measurement** with projector operators

$$P_m = |m\rangle\langle m|$$

Single qubit measurement in the $|+\rangle, |-\rangle$ basis

$$|+\rangle = H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad |-\rangle = H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

- A measurement of a qubit in the $|+\rangle, |-\rangle$ basis corresponds to perform a **projective measurement** with projectors

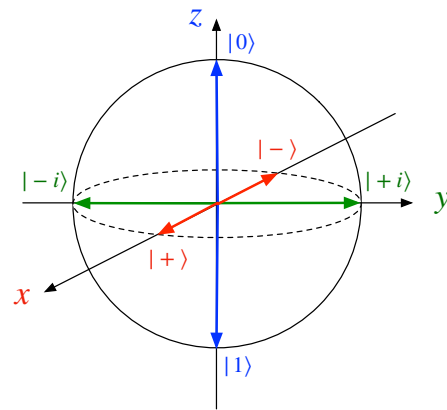
$$P_+ = |+\rangle\langle +| = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$P_- = |-\rangle\langle -| = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

- The completeness equation is satisfied:

$$P_+^\dagger P_+ + P_-^\dagger P_- = P_+ + P_- = I$$

Observable?



The Pauli matrix

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = |+\rangle\langle +| - |-\rangle\langle -|$$

$|+\rangle$ and $|-\rangle$ are the eigenvectors of X , with eigenvalues $+1$ and -1 , respectively.

Single qubit measurement in the $|+\rangle, |-\rangle$ basis

$$P_+ = |+\rangle\langle +| = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad P_- = |-\rangle\langle -| = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

A measurement of the state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ in the $|+\rangle, |-\rangle$ basis gives

- the result '+' with probability

$$p(+)=\langle\psi|P_+|\psi\rangle=\langle\psi|+\rangle\langle+|\psi\rangle=\frac{1}{2}|\alpha+\beta|^2$$

and the state after the measurement is $|+\rangle$

- the result '-' with probability

$$p(-)=\langle\psi|P_-|\psi\rangle=\langle\psi|-\rangle\langle-|\psi\rangle=\frac{1}{2}|\alpha-\beta|^2$$

and the state after the measurement is $|-\rangle$

Relative phase

The states $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

cannot be distinguished by measuring in the computational basis: both outcomes 0 and 1 occur with probability 1/2

Let us consider the $|+\rangle, |-\rangle$ basis, which has measurement operators

$$P_+ = |+\rangle\langle +| = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$P_- = |-\rangle\langle -| = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

If we measure the state $|+\rangle$ we always get the outcome '+'

$$p(+)=\langle +|P_+|+\rangle=\langle +|+\rangle\langle +|+\rangle=1$$

$$p(-)=\langle +|P_-|+\rangle=\langle +|-\rangle\langle -|+\rangle=0$$

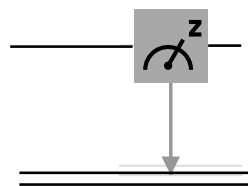
If we measure the state $|-\rangle$ we always get the outcome '-'

$$p(+)=\langle -|P_+|-\rangle=\langle -|+\rangle\langle +|-\rangle=0$$

$$p(-)=\langle -|P_-|-\rangle=\langle -|-\rangle\langle -|-\rangle=1$$

Measurement in the $|+\rangle, |-\rangle$ could distinguish the two states perfectly
→ relative phase matters!

Measurement with quantum circuits

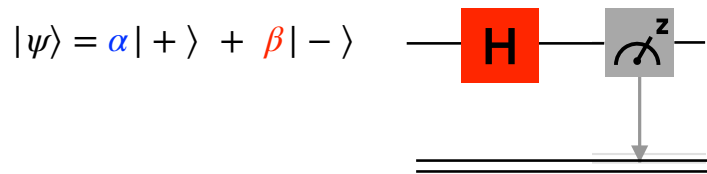


- For quantum processors **only measurements onto the computational basis are supported natively.**
- Measurements to other basis states can be performed by applying an appropriate unitary transformation before measurement.

Measurement with quantum circuits

How can we measure in the $|+\rangle, |-\rangle$ basis, or in any other orthonormal basis, with a quantum circuit?

First, we apply a gate to perform a basis change to the computational basis. Then a measurement is made in the computational basis



$$H|\psi\rangle = H(\alpha|+\rangle + \beta|-\rangle) = \alpha H|+\rangle + \beta H|-\rangle = \alpha|0\rangle + \beta|1\rangle$$

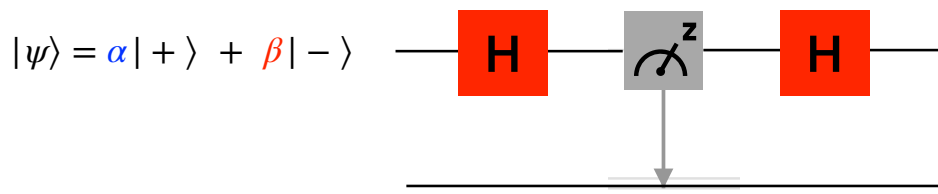
We measure **0** with probability $|\alpha|^2$ and **1** with probability $|\beta|^2$

The state of the system after this measurement is $|0\rangle$ or $|1\rangle$

How can we change back to the original basis, so that after the measurement we end up in the state $|+\rangle$ or in the state $|-\rangle$?

We perform the inverse basis change, by running the circuit backwards, replacing each gate by its inverse.

We use $H^{-1} (= H)$ to change back to the $|+\rangle, |-\rangle$ basis



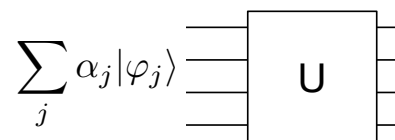
The state of the system after this measurement is now

- $|+\rangle$ with probability $|\alpha|^2$
- $|-\rangle$ with probability $|\beta|^2$

as expected for a measurement in the $|+\rangle, |-\rangle$ basis

$$\sum_j \alpha_j |\varphi_j\rangle$$

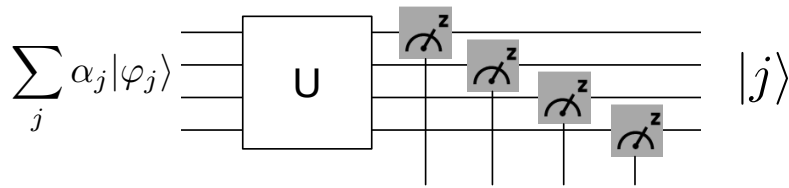
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$$U \left(\sum_j \alpha_j |\varphi_j\rangle \right) = \sum_j \alpha_j U |\varphi_j\rangle = \sum_j \alpha_j |j\rangle$$

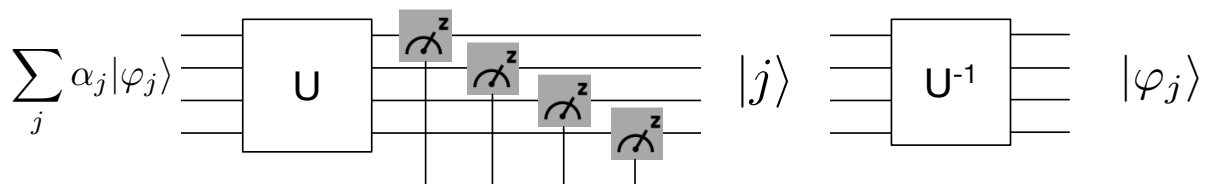


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- then, a measurement is made in the computational basis, obtaining a specific (classical) outcome with probability $|\alpha_j|^2$. The state of the system after this measurement is $|j\rangle$

- Finally U^{-1} is applied to change back to the $\{|\varphi_j\rangle\}$ -basis, leaving the post-measurement state $|\varphi_j\rangle$

The fourth postulate: Composition of Systems

Postulate 4: Composition of Systems Postulate

The state space of a **composite physical system** is the **tensor product** of the state spaces of the component subsystems. Moreover, if we have systems numbered 1 through n , and system number i is prepared in the state $|\psi_i\rangle$, then the joint state of the total system is

$$|\psi_1\rangle \otimes |\psi_2\rangle \otimes \dots \otimes |\psi_n\rangle$$

Note

- We often omit the $\otimes$ symbol and write the joint state like $|\psi_1\rangle |\psi_2\rangle \dots |\psi_n\rangle$, or sometimes even as $|\psi_1, \psi_2, \dots, \psi_n\rangle$ or $|\psi_1 \psi_2 \dots \psi_n\rangle$
- Why is the tensor product the mathematical structure used to describe the state space of a composite physical system?

Intuition: We need a new space which captures the interaction of all n systems

- **Not all states of a combined system can be separated into the tensor product of states of the individual components**

Computational basis: notation

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \quad |\varphi\rangle = \gamma|0\rangle + \delta|1\rangle = \begin{pmatrix} \gamma \\ \delta \end{pmatrix}$$

Joint state of the two qubits

$$|\psi\rangle \otimes |\varphi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \otimes \begin{pmatrix} \gamma \\ \delta \end{pmatrix} = \begin{pmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{pmatrix} = \alpha\gamma|00\rangle + \alpha\delta|01\rangle + \beta\gamma|10\rangle + \beta\delta|11\rangle$$

Standard basis notation

$$|00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad |01\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad |10\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad |11\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Separable states

A state $|\psi\rangle \in \mathbb{C}^n \otimes \mathbb{C}^m$ of a combined system is a **separable state** if it can be expressed as $|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$ for some $|\psi_1\rangle \in \mathbb{C}^n$ and $|\psi_2\rangle \in \mathbb{C}^m$

EXAMPLE

This two-qubit state is a **separable** state

$$\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle |+\rangle$$

Note

If the 2 qubits are prepared independently, and kept isolated, then each qubit forms a closed system, and the state **can** be written in the product form

Entangled states

A state of a composite system that **cannot** be written as a product of states of its component systems is called **entangled state**

Entangled states play a crucial role in quantum computation and quantum information

Note

Physical separation does not imply that the joint state must be separable
Two physically separated particles can be entangled (correlated)

Entangled states

EXAMPLE

The two-qubit state

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

cannot be written as a product of single-qubit states, hence it is **entangled**

Indeed, there are **NO** single qubit states $|\psi_1\rangle = \alpha|0\rangle + \beta|1\rangle$ and $|\psi_2\rangle = \gamma|0\rangle + \delta|1\rangle$ such that $|\psi\rangle = |\psi_1\rangle |\psi_2\rangle$

For all $\alpha, \beta, \gamma, \delta \in \mathbb{C}$

$$|\psi_1\rangle \otimes |\psi_2\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \otimes \begin{pmatrix} \gamma \\ \delta \end{pmatrix} = \begin{pmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{pmatrix} \neq \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Composite systems

Just as there are 2-qubit states that cannot be written as the product of two 1-qubit states, **there are 2-qubit gates (acting on both qubits) that cannot be written as a tensor product of two 1-qubit gates**

EXAMPLE: Quantum controlled-not, or CNOT gate

The CNOT gate flips the state of the second qubit if the first qubit is in state $|1\rangle$, and does nothing otherwise:

$$|00\rangle \rightarrow |00\rangle$$

$$|01\rangle \rightarrow |01\rangle$$

$$|10\rangle \rightarrow |11\rangle$$

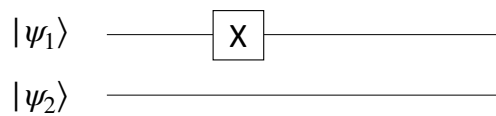
$$|11\rangle \rightarrow |10\rangle$$

The matrix representation for the CNOT gate **cannot** be written as a tensor product of two 1-qubit gates

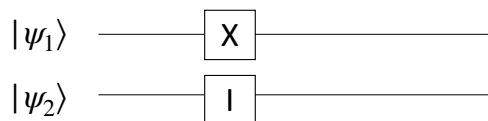
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Composite systems

Suppose we have a 2-qubit composite system $|\psi_1\rangle \otimes |\psi_2\rangle$, and we apply the NOT gate X only to the **first** qubit

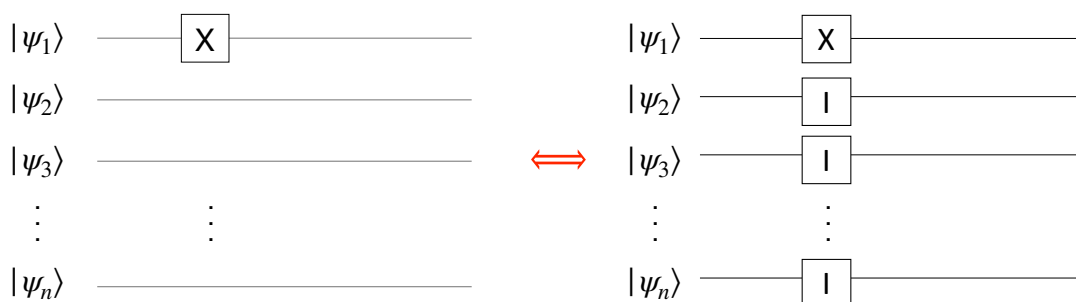


We implicitly apply the identity operator I to the **second** qubit at the same time



$|\psi_1\rangle \otimes |\psi_2\rangle$ gets mapped to $X|\psi_1\rangle \otimes I|\psi_2\rangle = (X \otimes I)(|\psi_1\rangle \otimes |\psi_2\rangle)$

Composite systems



For an n -qubit composite system $|\psi_1\rangle \otimes |\psi_2\rangle \otimes \dots \otimes |\psi_n\rangle$, applying the X gate to the first qubit corresponds to applying the operation

$$X \otimes I \otimes I \otimes \dots \otimes I$$

(with I repeated $n - 1$ times) to the entire system

Bell states

Four important two-qubit **entangled** states are the Bell states

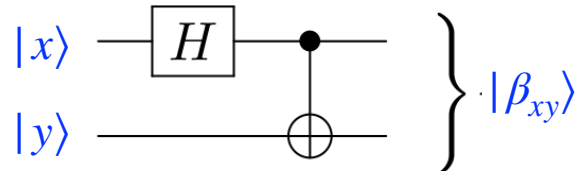
$$|\beta_{00}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

They form an orthonormal basis for $\mathbb{C}^4$

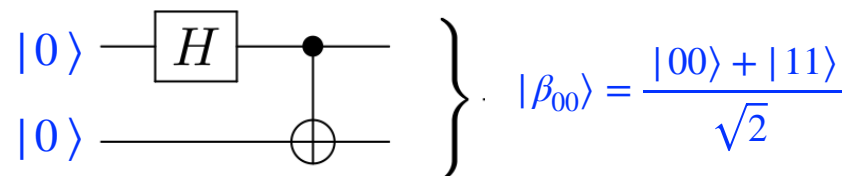
$$|\beta_{01}\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

$$|\beta_{10}\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}$$

$$|\beta_{11}\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}$$



Example



$$|0\rangle|0\rangle \xrightarrow{H \otimes I} \frac{|0\rangle + |1\rangle}{\sqrt{2}} |0\rangle = \frac{|0\rangle|0\rangle}{\sqrt{2}} + \frac{|1\rangle|0\rangle}{\sqrt{2}}$$

$$\xrightarrow{CNOT} \frac{|0\rangle|0\rangle}{\sqrt{2}} + \frac{|1\rangle|1\rangle}{\sqrt{2}} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Commonly used one-qubit and multi-qubit Quantum gate

One-qubit gates	Multi-qubit gates
Hadamard = $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	CNOT = $CX = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$
$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$	$CZ = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$
$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$	Controlled- $U = CU = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & U_{00} & U_{01} \\ 0 & 0 & U_{10} & U_{11} \end{pmatrix}$
NOT = $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	SWAP = $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$
$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	Toffoli (CCNOT) = $\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$
$R(\theta) = P(\theta) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}$	

Multiple qubits: measurement

Let us consider a two-qubit state

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

A complete set of measurement operators in the computational basis would be

$$P_{00} = |00\rangle\langle 00| \quad P_{01} = |01\rangle\langle 01| \quad P_{10} = |10\rangle\langle 10| \quad P_{11} = |11\rangle\langle 11|$$

When we measure the two qubits in the computational basis

- the probability of getting the result **00** is $|\alpha|^2$
- the probability of getting the result **01** is $|\beta|^2$
- the probability of getting the result **10** is $|\gamma|^2$
- the probability of getting the result **11** is $|\delta|^2$

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

How do we measure only the first qubit?

- If we measure the first qubit to be in the state $|0\rangle$, the measurement operator will consist of projectors onto the states $|00\rangle$ and $|01\rangle$, since **we must consider all states consistent with the measurement**

$$\begin{aligned} P_{0\star} &= |00\rangle\langle 00| + |01\rangle\langle 01| \\ &= (|0\rangle \otimes |0\rangle)(\langle 0| \otimes \langle 0|) + (|0\rangle \otimes |1\rangle)(\langle 0| \otimes \langle 1|) \\ &= |0\rangle\langle 0| \otimes |0\rangle\langle 0| + |0\rangle\langle 0| \otimes |1\rangle\langle 1| \\ &= |0\rangle\langle 0| \otimes (|0\rangle\langle 0| + |1\rangle\langle 1|) = |0\rangle\langle 0| \otimes I \end{aligned}$$

$$(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$$

- Similarly,

$$\begin{aligned} P_{1\star} &= |10\rangle\langle 10| + |11\rangle\langle 11| \\ &= (|1\rangle \otimes |0\rangle)(\langle 1| \otimes \langle 0|) + (|1\rangle \otimes |1\rangle)(\langle 1| \otimes \langle 1|) \\ &= |1\rangle\langle 1| \otimes |0\rangle\langle 0| + |1\rangle\langle 1| \otimes |1\rangle\langle 1| \\ &= |1\rangle\langle 1| \otimes (|0\rangle\langle 0| + |1\rangle\langle 1|) = |1\rangle\langle 1| \otimes I \end{aligned}$$

We tensor the desired measurement operator with the identity matrix

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

The probability of measuring the first qubit in state $|0\rangle$ is therefore

$$\begin{aligned} p(0) &= \langle \psi | P_{0\star} | \psi \rangle = \langle \psi | |0\rangle\langle 0| \otimes I | \psi \rangle \\ &= (\alpha^* \ \beta^* \ \gamma^* \ \delta^*) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{pmatrix} = (\alpha^* \ \beta^* \ \gamma^* \ \delta^*) \begin{pmatrix} \alpha \\ \beta \\ 0 \\ 0 \end{pmatrix} = |\alpha|^2 + |\beta|^2 \end{aligned}$$

$p(0)$ is the sum of the probabilities of measuring $|00\rangle$ and $|01\rangle$, as expected

The state after the measurement is

$$\begin{aligned} \frac{P_{0\star} |\psi\rangle}{\sqrt{p(0)}} &= \frac{1}{\sqrt{|\alpha|^2 + |\beta|^2}} (|0\rangle\langle 0| \otimes I) (\alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle) \\ &= \frac{1}{\sqrt{|\alpha|^2 + |\beta|^2}} (\alpha|00\rangle + \beta|01\rangle) \end{aligned}$$

The relative probability of observing $|00\rangle$ and $|01\rangle$ hasn't changed, but by measuring $|0\rangle$ on the first qubit, we have eliminated the possibility that the system is in state $|10\rangle$ or $|11\rangle$

Example

$$|\psi\rangle = \sqrt{\frac{1}{11}}|00\rangle + \sqrt{\frac{5}{11}}|01\rangle + \sqrt{\frac{2}{11}}|10\rangle + \sqrt{\frac{3}{11}}|11\rangle$$

When we measure the two qubits in the computational basis,

- the probability of getting the result **00** is **1/11**
- the probability of getting the result **01** is **5/11**
- the probability of getting the result **10** is **2/11**
- the probability of getting the result **11** is **3/11**

The probability of measuring 0 in the first qubit is **1/11+5/11= 6/11**

The state after measuring 0 in the first qubit becomes

$$\sqrt{\frac{11}{6}}\left(\sqrt{\frac{1}{11}}|00\rangle + \sqrt{\frac{5}{11}}|01\rangle\right)$$

- the state of the first qubit is $|0\rangle$
- the state of the second qubit is left in superposition